Chirag Hegde

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SKILLS

- Languages: Python, R, C, Java, CSS, HTML, JavaScript, NextJS, ReactJS, Kotlin, Tableau, Ruby, Vue.js, Chart.js
- Database and Operating Systems: PostgreSQL, MySQL, MongoDB, Ubuntu
- Tools/Frameworks: GIT, Postman API, AWS, Hadoop, Docker, Swagger
- Certifications: Certified as an AWS Academy Graduate - AWS Academy Cloud Foundations from ICT Academy, Palo Alto Networks Academy Cybersecurity Foundation (Coursera), HTML, CSS, and JavaScript for Web Developers (Coursera)

PROFESSIONAL EXPERIENCE

AI Software Engineer, Cloud Pharmaceuticals - Durham, NC USA *June 2025 – Present*

- Prototyped an LLM-based pipeline to extract structured data from Epic EMR notes (demographics, conditions, medications) and normalize to FHIR with RxNorm/ICD-10 mapping.
- Assembled longitudinal patient timelines capturing medication start/stop, dose changes, encounters, and outcomes.
- Trained and evaluated baseline models and rules in Python (scikit-learn, med-spaCy).

Teaching Assistant for CSC 565*:* Graph Theory - NCSU *Aug 2024 – May 2025*

- Supported graduate-level instruction by assisting with coursework design, grading assignments, and conducting problem-solving sessions.
- Held weekly office hours to help students master topics such as graph traversal, connectivity, planarity, and network flows.

Software Engineer Intern, **TetherFi**, - Bengaluru, India *May 2022 – Nov 2022*

- Developed an MVC Asp.net Core web application for an HRIS system focused on recruitment, leveraging Entity Framework for ORM capabilities and integrating Identity for user authentication as part of the core product development team.
- Engineered a chatbot that retrieves institutional documents from any server within the organization, utilizing RESTful APIs for data fetching and JSON for data interchange to enhance precision and efficiency.
- Designed an advanced NLP model using TensorFlow for machine learning algorithms and BERT for natural language understanding, significantly improving the chatbot's ability to provide accurate and contextually relevant document retrieval.

Software Engineer Intern, **SoftLink -** Mumbai, India *Feb* 2021*– Dec 2021*

- Supported the successful implementation of a digital transformation initiative, utilizing Azure cloud services and SAP ERP solutions to overhaul logistics operations.

PROJECTS

Student Attentiveness Evaluation in Classroom Using Face Recognition and Machine Learning

- Developed a real-time attentiveness checker using facial cues (head pose, gaze direction) and student-specific CNN models to classify engagement.
- Built a browser-based frontend and Flask backend where OpenCV captures frames and TensorFlow classifies attention at 30 FPS, with results stored in MongoDB.
- Technologies used: Python, Flask, OpenCV, TensorFlow, Google OAuth API, MongoDB, JavaScript

E-Commerce Storefront with Stripe Integration

- Built a full-featured e-commerce platform for digital product sales.
- Developed a responsive, user-friendly UI and implemented seamless payment processing with Stripe for secure transactions.
- Integrated secure payment flows using the Stripe API, supporting one-time and recurring payments.
- Technologies used: Next.js, TypeScript, Stripe API

Sign Language Detection Using Deep Learning

- Utilized deep learning to recognize and interpret sign language with 84% accuracy using a Convolutional Neural Network (CNN) optimized through hyperparameter tuning.
- Technologies used: Python, TensorFlow, CNN, Kaggle datasets

PersonalDoctor LLM based Healthcare

- Built a chatbot that answers medical questions and suggests likely conditions from symptoms using a fine-tuned RoBERTa model (Q&A) and a Random Forest classifier (symptom check).
- Collected and cleaned data by scraping trusted health sites to create a 6.8k Q&A set; prepared a Kaggle disease–symptom table for training.
- Trained and evaluated the models with Hugging Face/fastai/PyTorch and scikit-learn
- Deployed a lightweight web demo with Flask and Gradio so users can ask questions and see predictions in real time.
- Technologies used: Python, HuggingFace Transformers (RoBERTa), Fast.ai, PyTorch, scikit-learn , Flask, Gradio

EDUCATION

North Carolina State University - Raleigh, United States *Aug 2023 - May 2025*

- *Master of Computer Science* ***GPA: 3.63/4.0***

- Coursework: Automated Learning and Data Analysis, Artificial Intelligence, Social Computing and Decentralized AI, Graph Theory, Cloud Computing, Software Engineering, Neural Network and Deep Learning, Object-oriented Design and Development, Design and Analysis of Algorithms, Parallel Systems

A.P. Shah Institute of Technology - Thane, India *Aug 2019 - May 2023*

- *Bachelor of Engineering in Computer Engineering* ***CGPA :8.9/10***

- Coursework: Data Analytics, Machine Intelligence, AI, Cloud Computing, Computer Networks, OS, Big Data, Network Analysis and Mining, Compiler Design, Microprocessors, Software and Systems Performance